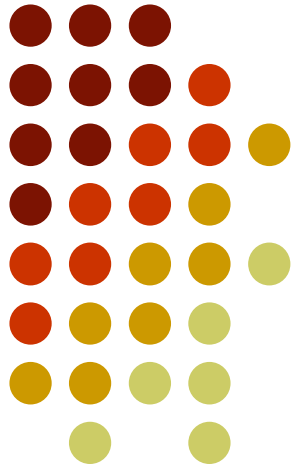
Interest Rates

Chapter 4



Types of Rates



- Treasury rates
- LIBOR rates
- Repo rates

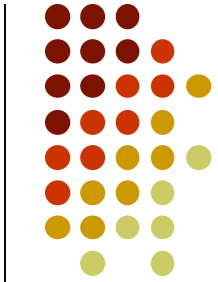
Treasury Rates

- Rates on instruments issued by a government in its own currency

LIBOR and LIBID

- LIBOR is the rate of interest at which a bank is prepared to deposit money with another bank. (The second bank must typically have an AA rating)
- LIBOR is compiled once a day by the British Bankers Association on all major currencies for maturities up to 12 months
- LIBID is the rate at which an AA bank is prepared to pay on deposits from anther bank

Repo Rates



- Repurchase agreement is an agreement where a financial institution that owns securities agrees to sell them today for X and buy them back in the future for a slightly higher price, Y
- The financial institution obtains a loan.
- The rate of interest is calculated from the difference between X and Y and is known as the repo rate

The Risk-Free Rate

- The short-term risk-free rate traditionally used by derivatives practitioners is LIBOR
- The Treasury rate is considered to be artificially low for a number of reasons (See Business Snapshot 4.1)

Measuring Interest Rates



- The compounding frequency used for an interest rate is the unit of measurement
- The difference between quarterly and annual compounding is analogous to the difference between miles and kilometers

Impact of Compounding

When we compound m times per year at rate R an amount A grows to $A(1+R/m)^m$ in one year

Compounding frequency	*Value of $100 in one year at 10%*
Annual (m=1)	110.00
Semiannual (m=2)	110.25
Quarterly (m=4)	110.38
Monthly (m=12)	110.47
Weekly (m=52)	110.51
Daily (m=365)	110.52

Continuous Compounding

(Pages 84-85)



- In the limit as we compound more and more frequently we obtain continuously compounded interest rates
- \$100 grows to $\$100e^{RT}$ when invested at a continuously compounded rate R for time T
- \$100 received at time T discounts to $\$100e^{-RT}$ at time zero when the continuously compounded discount rate is R

Conversion Formulas

(Page 85)



Define

R_c: rate with continuous compounding

R_m: equivalent rate with compounding m times per year

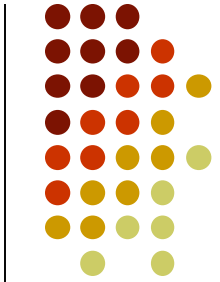
$$R_c = m \ln\left(1 + \frac{R_m}{m}\right)$$

$$R_m = m\left(e^{R_c / m} - 1\right)$$

Examples

- 10% with semiannual compounding is equivalent to $2\ln(1.05)=9.758\%$ with continuous compounding
- 8% with continuous compounding is equivalent to $4(e^{0.08/4}-1)=8.08\%$ with quarterly compounding
- Rates used in option pricing are nearly always expressed with continuous compounding

Zero Rates



A zero rate (or spot rate), for maturity T is the rate of interest earned on an investment that provides a payoff only at time T

Example (Table 4.2, page 87)

Maturity (years)	Zero Rate (% cont. comp.)
0.5	5.0
1.0	5.8
1.5	6.4
2.0	6.8

Bond Pricing



- To calculate the cash price of a bond we discount each cash flow at the appropriate zero rate
- In our example, the theoretical price of a two-year bond providing a 6% coupon semiannually is

$$3e^{-0.05\times 0.5} + 3e^{-0.058\times 1.0} + 3e^{-0.064\times 1.5} + 103e^{-0.068\times 2.0} = 98.39$$

Bond Yield

- The bond yield is the discount rate that makes the present value of the cash flows on the bond equal to the market price of the bond
- Suppose that the market price of the bond in our example equals its theoretical price of 98.39
- The bond yield is given by solving

$$3e^{-y\times 0.5} + 3e^{-y\times 1.0} + 3e^{-y\times 1.5} + 103e^{-y\times 2.0} = 98.39$$

to get y = 0.0676 or 6.76% with cont. comp.

Par Yield

- The par yield for a certain maturity is the coupon rate that causes the bond price to equal its face value.
- In our example we solve

$$\frac{c}{2}e^{-0.05\times 0.5}+\frac{c}{2}e^{-0.058\times 1.0}+\frac{c}{2}e^{-0.064\times 1.5}$$
$$+\left(100+\frac{c}{2}\right)e^{-0.068\times 2.0}=100$$

to get $c=6.87$ (with s.a. compounding)

Par Yield continued

In general if m is the number of coupon payments per year, d is the present value of $1 received at maturity and A is the present value of an annuity of $1 on each coupon date

$$c = \frac{(100 - 100d)m}{A}$$

(in our example, m = 2, d = 0.87284, and A = 3.70027)

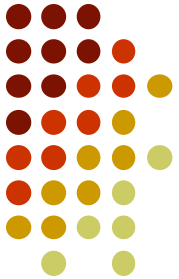
Data to Determine Zero Curve

(Table 4.3, page 88)

Bond Principal	Time to Maturity (yrs)	Coupon per year ($)*	Bond price ($)
100	0.25	0	97.5
100	0.50	0	94.9
100	1.00	0	90.0
100	1.50	8	96.0
100	2.00	12	101.6

* Half the stated coupon is paid every six months.

The Bootstrap Method



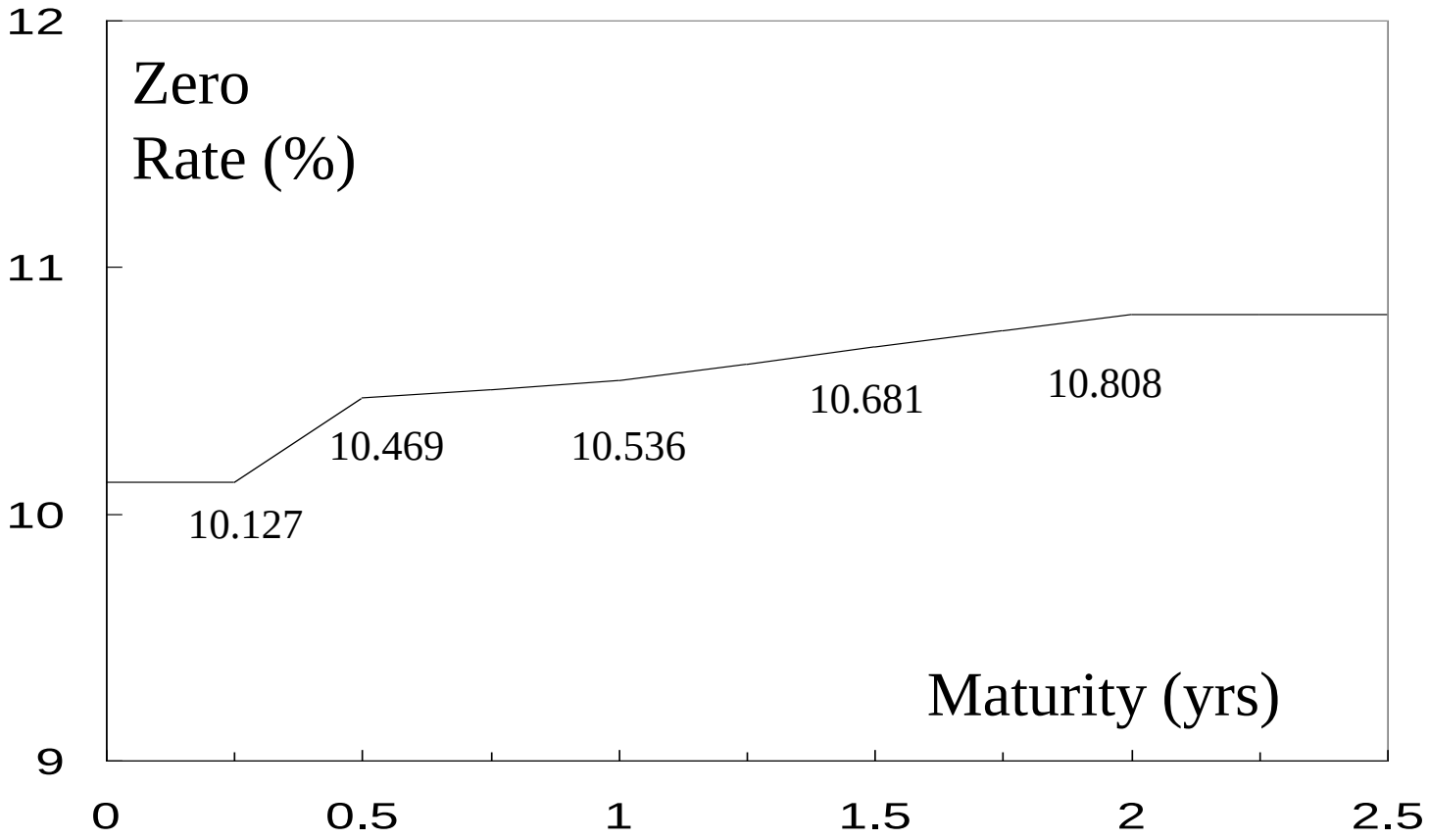
- This is 10.127% with continuous compounding
- Similarly the 6 month and 1 year rates are 10.469% and 10.536% with continuous compounding
- To calculate the 1.5 year rate we solve

$$4e^{-0.10469\times 0.5} + 4e^{-0.10536\times 1.0} + 104e^{-R\times 1.5} = 96$$

to get R = 0.10681 or 10.681%

- Similarly the two-year rate is 10.808%

Zero Curve Calculated from the Data (Figure 4.1, page 89)



Forward Rates

The forward rate is the future zero rate implied by today's term structure of interest rates

Formula for Forward Rates

- Suppose that the zero rates for time periods T_1 and T_2 are R_1 and R_2 with both rates continuously compounded.
- The forward rate for the period between times T_1 and T_2 is

$$\frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

- This formula is only approximately true when rates are not expressed with continuous compounding

Application of the Formula

Year (n)	Zero rate for n-year investment (% per annum)	Forward rate for nth year (% per annum)
1	3.0	
2	4.0	5.0
3	4.6	5.8
4	5.0	6.2
5	5.5	6.5

Upward vs Downward Sloping Zero Curve



- For an upward sloping zero curve:

Fwd Rate > Zero Rate

- For a downward sloping zero curve

Zero Rate > Fwd Rate

Discussion

- The term structure of interest rates is upward sloping. Put the following in order of magnitude:
  a) The five-year zero rate
  b) The yield on a five-year coupon-bearing bond
  c) The forward rate corresponding to the period between 4.75 and 5 years in the future

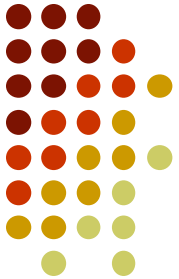
  What is the answer to this question when the term structure of interest rates is downward sloping?

Forward Rates (Con't)



- By borrowing and lending at zero rates, a large financial institution can lock in forward rates.
  - Borrow $100 at 3% for one year and invest the money at 4% for two years
  - Borrow $100 for four years at 5% and invest it for three years at 4.6%

Forward Rate Agreement



- A forward rate agreement (FRA) is an agreement that a certain rate will apply to a certain principal during a certain future time period
  - R_K: The rate of interest agreed to in the FRA
  - R_F: the forward LIBOR interest rate for the period between T_1 and T_2, calculated today
  - R_M: The actual LIBOR interest rate observed in the market at time T_1 for the period between T_1 and T_2
  - L: The principal underlying the contract

Forward Rate Agreement (Con't)



- Consider an FRA where company X agrees to lend money to company Y for the period of time between T_1 and T_2
  - Cash flow to X at T_2: $L(R_K - R_M)(T_2 - T_1)$
  - Cash flow to Y at T_2: $L(R_M - R_K)(T_2 - T_1)$
- Usually FRAs are settled at T_1 rather than T_2
  - X's payoff at T_1: $\frac{L(R_K - R_M)(T_2 - T_1)}{1 + R_M(T_2 - T_1)}$
  - Y's payoff at T_1: $\frac{L(R_M - R_K)(T_2 - T_1)}{1 + R_M(T_2 - T_1)}$

Example: Cash Flows from an FRA



- Suppose that a company enters into an FRA that is designed to ensure it will receive a fixed rate of 4% on a principal of $100 million for a three-month period starting in three years. The three-month LIBOR proves to be 4.5% for the three-month period.

Forward Rate Agreement: Valuation



- The value of an FRA is always zero when $R_K = R_F$.
- Compare two FRAs.
  - LIBOR forward rate R_F will be received
  - R_K will be received
- With an interest rate of R_K, the interest cash flow is $R_K(T_2 - T_1)$ at time T_2
- With an interest rate of R_F, the interest cash flow is $R_F(T_2 - T_1)$ at time T_2

Valuation Formulas (con't)



- When the rate R_K will be received on a principal of L the value of the FRA is the present value of

$$(R_K - R_F)(T_2 - T_1)$$

received at time T_2, i.e.,

$$V_{FRA} = L(R_K - R_F)(T_2 - T_1)e^{-R_2 T_2}$$

- When the rate R_K will be paid on a principal of L the value of the FRA is the present value of

$$(R_F - R_K)(T_2 - T_1)$$

received at time T_2, i.e.,

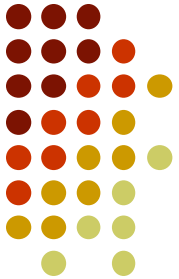
$$V_{FRA} = L(R_F - R_K)(T_2 - T_1)e^{-R_2 T_2}$$

Valuation Example

- Zero and forward rates are as in the table of Slide 23
- An FRA entered into some time ago ensures that a company will receive 6%, measured with annual compounding, on $100 million between the end of year 1 and the end of year 2
- Forward LIBOR is 5% with continuous compounding or 5.127% with annual compounding.
- The value of the FRA (in $ millions) is

$$100\times(0.06-0.05127)\times e^{-0.04\times 2}=0.8058$$

Valuation Formulas (con't)



- An FRA can be valued if we
  - Calculate the payoff on the assumption that forward rates are realized (that is, $R_M = R_F$)
  - Discount the payoff at the risk-free rate

Theories of the Term Structure



- Expectations Theory: forward rates equal expected future zero rates
- Market Segmentation: short, medium and long rates determined independently of each other
- Liquidity Preference Theory: forward rates higher than expected future zero rates

Liquidity Preference Theory

- Suppose you have been offered the following choices

Maturity	Deposit rate	Mortgage rate
1 year	3%	6%
5 year	3%	6%

- What would you choose as a depositor? What for your mortgage?

Liquidity Preference Theory cont

- To match the maturities of borrowers and lenders a bank has to increase long rates above expected future short rates
- In our example the bank might offer

Maturity	Deposit rate	Mortgage rate
1 year	3%	6%
5 year	4%	7%

Summary

- Compounding
  - Conversion formulas

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right)$$

$$R_m = m\left(e^{R_c / m} - 1\right)$$

Summary

- Zero rates and Forward rates
  - Zero curve determination: bootstrapping
  - Forward rate for the period between T_1 and T_2

$$\frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

  - Upward sloping zero cure:
    - Fwd rate ___ Zero rate
  - Downward sloping zero cure:
    - Fwd rate ___ Zero rate

Summary

- Bond pricing, bond yield, and par yield
  - Bond pricing: discount each cash flow at the appropriate zero rate
  - Bond yield: discount rate that makes
    - the PV(cash flows) = market price of the bond
  - Par yield: coupon rate that causes
    - Bond price = face value

$$c = \frac{(100 - 100d)m}{A}$$

Summary

- Forward rate agreement: an agreement that a certain rate will apply to a certain principal during a certain future time period
  - Payoff
  - Valuation
- Theories of the term structure
  - Liquidity preference theory: forward rates higher than expected future zero rates